

Summary

Cell calling

Duplication and PCR quality

Library saturation and top-up estimate

Recommendation

scRNA-seq QC Report

My Sample | Human
20 April 2026

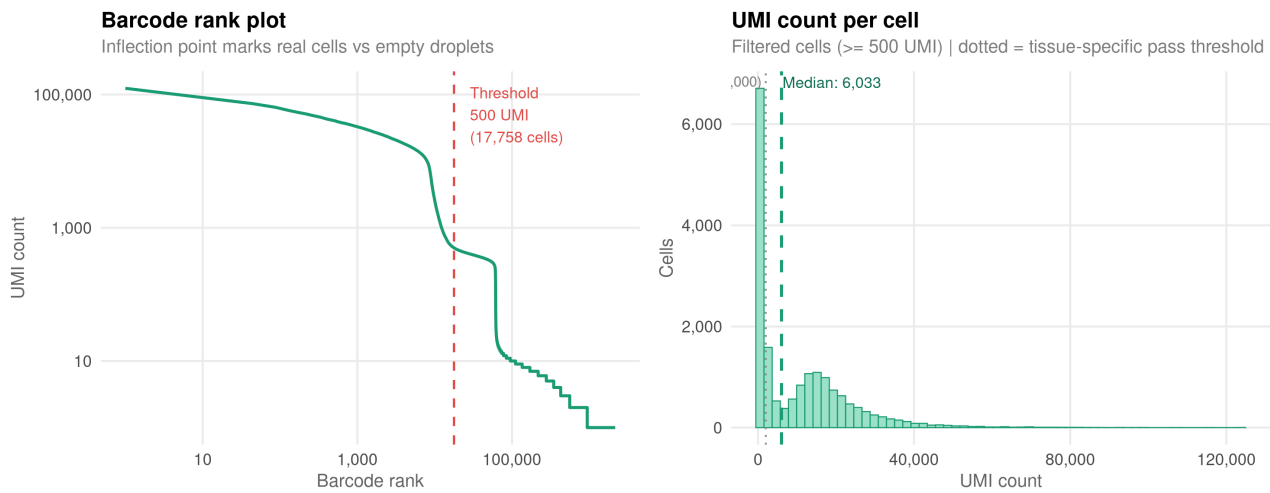
Summary

Metric	Value	Status
Sample	My Sample	
Species	Human	
Tissue type	general	
UMI threshold applied	500	
Cells called	17,758	PASS
Median UMI / cell	6,033	PASS
Mean UMI / cell	10,803	
Mean duplication rate	35.4%	SHALLOW
Overall duplication rate	36.3%	
Est. library size / cell	9,402	
Est. unseen UMI / cell	3,369	
Fraction of library seen	36.3%	

Tissue context: General defaults for mixed or unlisted tissue types.
UMI thresholds for general : PASS >= 2,000 | REVIEW >= 800

Top-up sequencing: RECOMMENDED
At 35.4% mean duplication, only 36.3% of the estimated library has been observed. A 1.8x top-up run on the existing library is recommended to reach ~50% duplication and ~6,005 median UMI/cell.

Cell calling

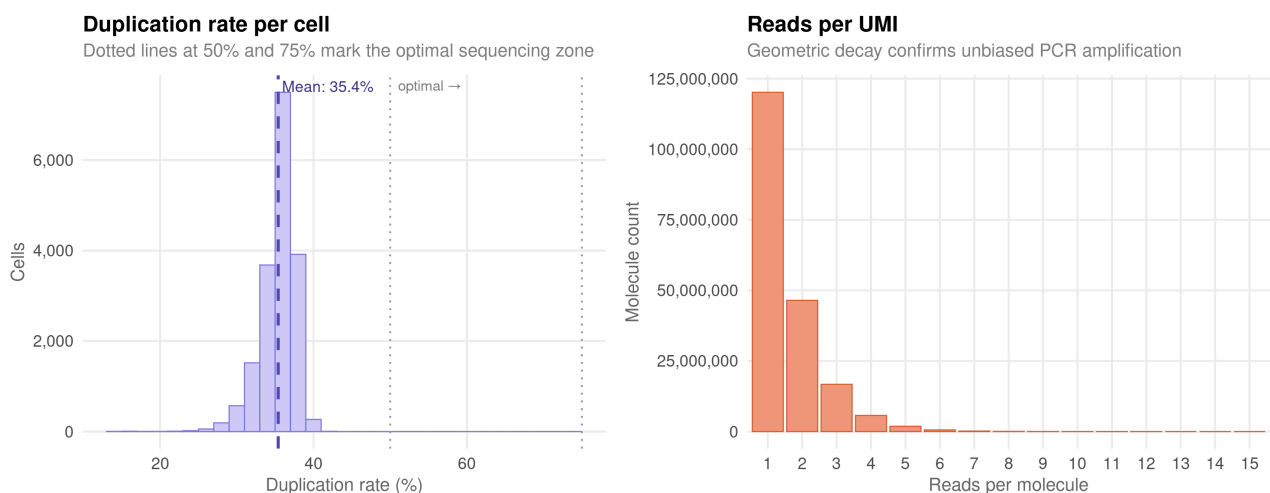


The barcode rank plot shows UMI counts from highest to lowest. The inflection separates real cells from empty droplets. A threshold of **500 UMI** was applied, yielding **17,758 cells**.

Cell count and UMI statistics at each threshold

UMI threshold	Cells	Median UMI	Mean UMI
100	61,326	395.0	3,387
200	60,987	395.0	3,405
500	17,758	6,033.0	10,803
1,000	12,275	14,009.0	15,339
2,000	10,464	15,711.5	17,751
5,000	9,075	17,132.0	19,989

Duplication and PCR quality

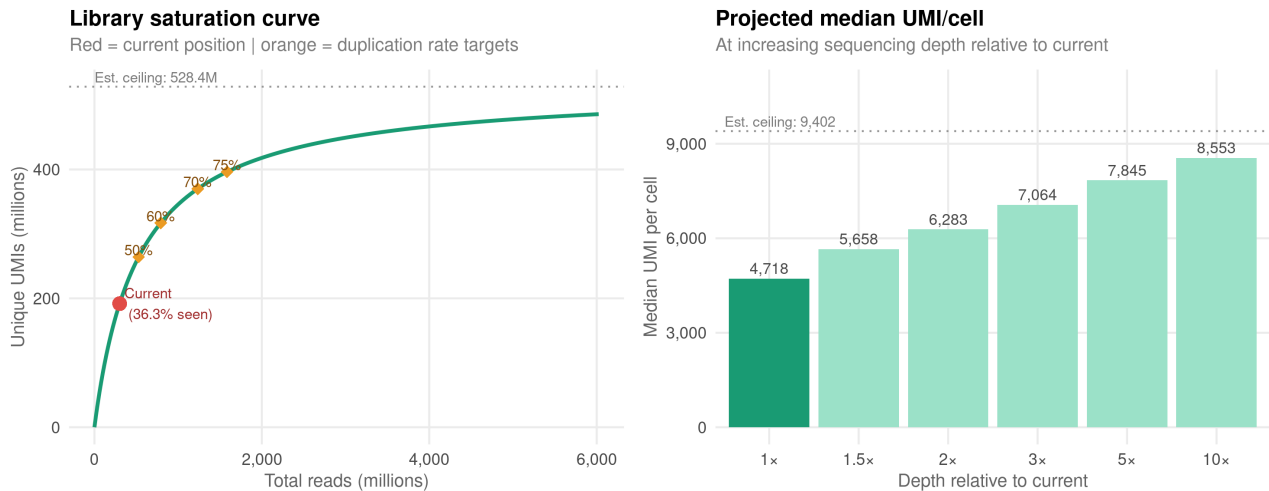


The **duplication rate** ($1 - \text{unique UMIs} / \text{total reads}$) measures the fraction of reads that are PCR copies of molecules already counted. A narrow, symmetric distribution indicates uniform sequencing depth and unbiased PCR amplification. The optimal zone (50-75%) is marked by dotted

lines.

The **reads-per-UMI** plot shows the molecule-level read support distribution. A geometric decay (each bar ~3x smaller than the previous) confirms no PCR jackpotting or amplification bias.

Library saturation and top-up estimate



Library size is estimated using the Michaelis-Menten model:

$\text{est_library} = \text{observed_UMIs} / \text{duplication_rate}$. This assumes uniform sampling probability — highly expressed genes are near-complete while rare transcripts make up most of the unseen fraction.

Reads required to reach each duplication rate target

Target dup rate	Total reads needed (M)	Fold increase	Projected median UMI/cell
50%	528	1.8	6005
60%	793	2.6	6827
70%	1,233	4.1	7567
75%	1,585	5.3	7910

Recommendation

Top-up sequencing recommended

Current duplication rate: 35.4% (below the 50% optimal threshold).

Implication: The library is under-sequenced relative to its complexity. Only 36.3% of the estimated 528,434,951 molecule library has been observed.

Library quality: The narrow duplication rate distribution and geometric reads-per-UMI decay confirm this is a sequencing depth issue — not a library preparation or PCR quality issue.

Action: Request a **1.8x top-up run** on the existing library. No new library preparation is needed.

Target dup rate	Total reads needed (M)	Fold increase	Projected median UMI/cell
50%	528	1.8x	6,005
60%	793	2.6x	6,827
70%	1,233	4.1x	7,567
75%	1,585	5.3x	7,910

Note on the saturation model: Estimates assume equal sampling probability across all molecules. In reality, highly expressed genes are already near-completely sampled; the unseen fraction (3,369 UMI/cell) is enriched for lowly expressed transcripts. Top-up sequencing disproportionately benefits detection of rare or lowly expressed genes.

Generated by *scrna_qc_report.R* | 2026-04-20 11:57

